

STROKE RISK DETECTION SYSTEM USING MACHINE LEARNING

Project Report

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Abstract

This project develops a Stroke Risk Detection System using the Symptoms.csv dataset from Kaggle. Data preprocessing included handling missing values using `fillna()`, encoding categorical variables using `get_dummies()`, balancing the dataset with SMOTE, splitting data into training and testing sets, and training a Random Forest Classifier. The model was evaluated using accuracy metrics to identify stroke risk.

Introduction

Stroke is a major health concern worldwide. Early identification of stroke risk can help healthcare professionals take preventive measures. This project applies machine learning techniques to detect individuals who may be at risk of stroke.

Objectives

Analyze symptom data, preprocess the dataset, balance classes using SMOTE, train a Random Forest model, evaluate accuracy, and detect stroke risk.

Dataset

Dataset Used: Symptoms.csv (Kaggle)

Methodology

Data cleaning using `fillna()`, feature encoding using `get_dummies()`, train-test splitting, SMOTE balancing, Random Forest training, prediction, and accuracy evaluation.

Results

The model successfully detected stroke risk patterns from the dataset and achieved reliable performance based on accuracy evaluation.